# 71ST KU STRUCTURAL ENGINEERING ANNUAL CONFERENCE

# Longitudinal Housing Recovery After Hurricane Matthew and Implications for Disaster Resilience

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## Background and motivation

Long-term housing recovery after disasters is **uneven and non-linear**, but most studies observe only damage or short-term recovery. This study uniquely tracked the **same housing units** in Lumberton, NC for seven years after Hurricane Matthew to characterize long-term recovery trajectories.

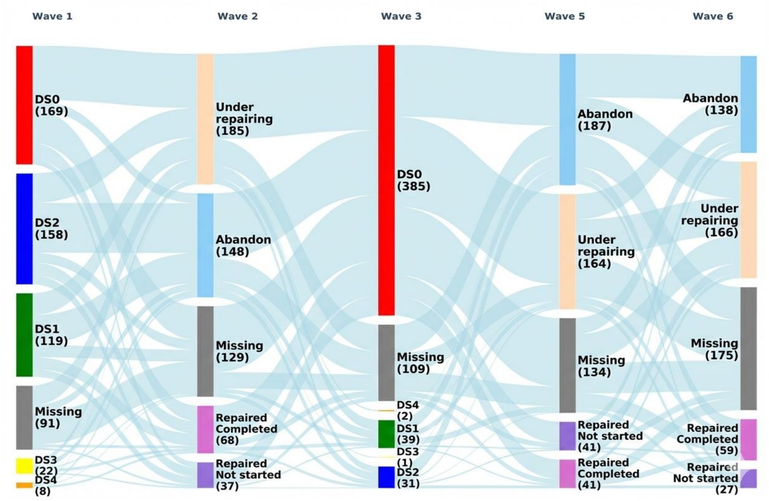



**Figure 1:** Tracking Housing Damage and Recovery Trajectories Over Five Field Studies

## Study area and data

**Where**: Lumberton, North Carolina

**Who**: 568 housing units sampled through a stratified cluster design with 3:1 oversampling in high-probability flooded areas

**When**: nine data collection trips from 2016–2023

**How:** we used closed-ended surveys capturing differences in impact, damage, recovery progress, and adaptation strategies.

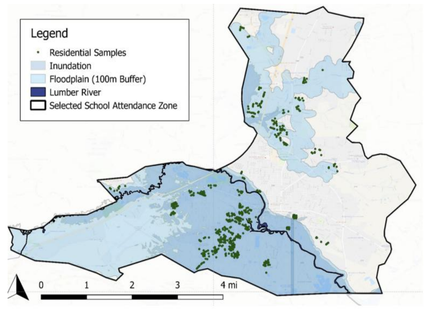



**Figure 2:** Sampled Housing Units

## Timeline of Hazards & Survey Waves

This study began shortly after **Hurricane Matthew (Oct 2016).** It continued **Hurricane Florence (Sept 2018)** and through the **COVID-19 pandemic**. The field study concluded with a final field study conducted in October 2023.

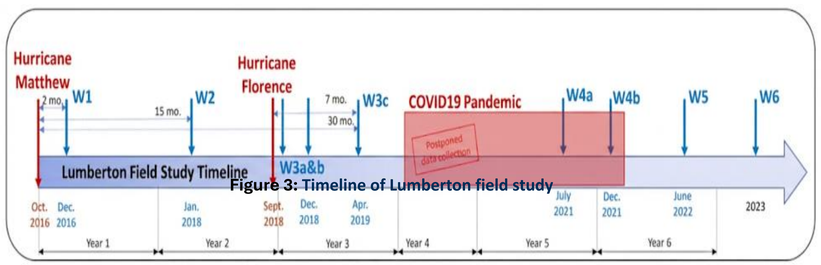
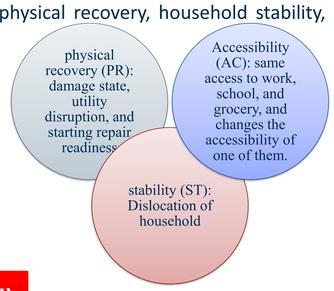



**Figure 3:** Timeline of Lumberton field study

## Longitudinal Housing Recovery Metric

Housing recovery is measured using the Longitudinal Household Recovery Capital (LHRC), a multidimensional index that captures physical recovery, household stability, and housing accessibility.

LHRC integrates damage, displacement, and access to essential services into a single score that can be tracked longitudinally for the same housing units.




$$LHRC_{ij} = f(PR_{ij}, ST_{ij}, AC_{ij})$$

$$LHRC_{ij} = \frac{3\,PR_{ij} + 1\,ST_{ij} + 2\,AC_{ij}}{6}$$

## LHRC patterns over time (overall trend)

Changes in LHRC over time indicate that recovery is **not monotonic** but unfolds unevenly, with periods of improvement, stagnation, and occasional decline occurring across different waves.

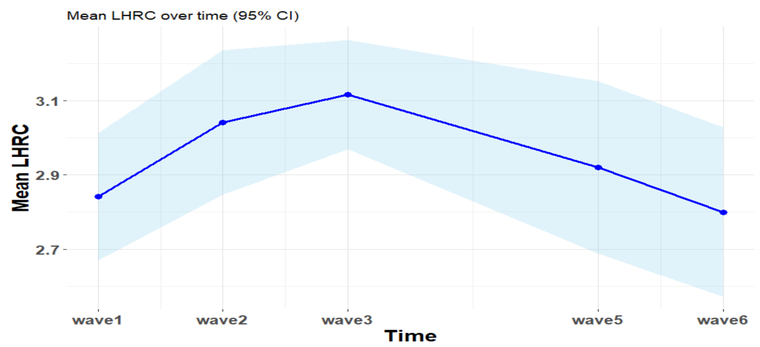



**Figure 4:** Mean LHRC over waves with 95% CI

## Trajectory types (DTW clustering)

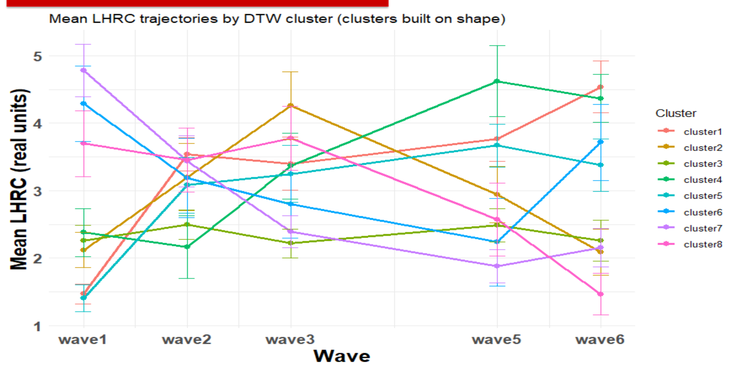



**Figure 5:** Mean LHRC trajectories captured in 7 DTW clusters

## Pre-disaster insurance coverage

### Flood Insurance

- Flood insurance is rare in the sample (**24.5% Yes**,**73.1% No**, **2.3% DK**).
- **Cluster 7** had highest prevalence of flood insurance (18 households; 6.0% of the total sample.

### Homeowner's Insurance

- Homeowner's insurance was more **common** (**58.0% Yes**, **39.6% No**, **2.6% DK**).
- **Cluster 3** (**13.2%**), **Cluster 7** (**9.2%**) and **Cluster 1** (**8.0%**) had highest prevalence of homeowner's insurance.

## Model specification

$$LHRC_{it} = \beta_0 + \beta_1(FloodIns_i) + \beta_2(HomeIns_i) + \beta_3(Wave_t) + \beta_4(Insurance \times Wave) + b_i + \varepsilon_{it}$$

Reference Group: when Flood Insurance =Yes and Home Insurance = Yes at Wave 1 Intercept ($\beta_0$ ) = the predicted mean for the reference group at the reference wave.

| Term | numDF | denDF | F-value | p-value |
|---|---|---|---|---|
| (Intercept) | 1 | 605 | 1593.7171 | <0.0001 |
| floodinsurance | 1 | 243 | 1.1999 | 0.2744 |
| homeinsurance | 1 | 605 | 7.1916 | 0.0075 |
| wave_f | 4 | 605 | 2.2897 | 0.0585 |
| floodinsurance:wave_f | 4 | 605 | 1.2210 | 0.3006 |
| homeinsurance:wave_f | 4 | 605 | 0.3488 | 0.8449 |

## Insurance and longitudinal recovery patterns

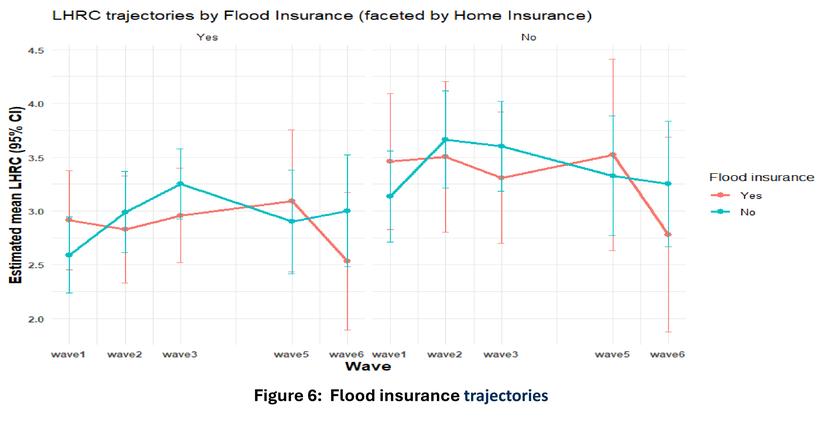



**Figure 6: Flood insurance trajectories**

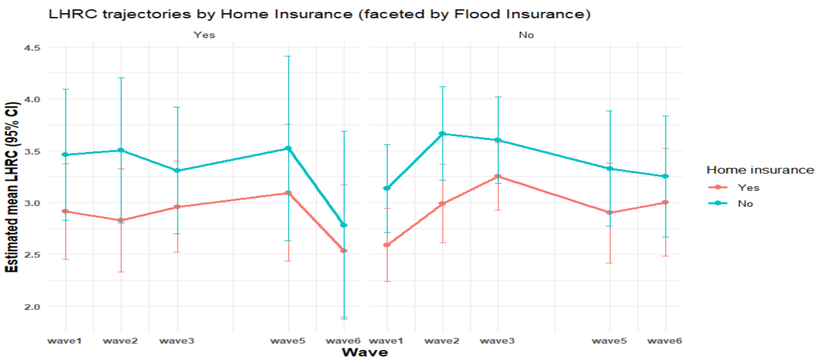



**Figure 7: Homeowner's insurance trajectories**

## Preliminary Conclusions

- Insurance status affects average LHRC level but does not strongly change the shape of recovery trajectories over time.
- These longitudinal patterns highlight substantial heterogeneity in long-term housing recovery that would not be visible from single-wave assessments.

## Acknowledgements

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